

Homework 34: Curve Fitting Using Least Squares Problem

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BCS A, E, F, G

Problem 1. Let $(1,0)$, $(2,1)$, and $(3,3)$ be three points in $\mathbb{R}^2$. How can you find the line $y = c_0 + c_1x$ that “best fits” these points? Sketch it.

Problem 2. Fit a quadratic function to the four data points $(a_1, b_1) = (-1,8)$, $(a_2, b_2) = (0,8)$, $(a_3, b_3) = (1,4)$, and $(a_4, b_4) = (2,16)$. Sketch it.

Problem 3. Find a cubic polynomial whose graph passes through the points $(1,3)$, $(-1,13)$, $(2,1)$, $(-2,33)$. Sketch it.

Problem 4. Find the linear and quadratic regression for the data points $(8,1)$, $(0,2)$, $(10,4)$, $(10,5)$, $(2,6)$. Sketch it.

Note: You can use “Desmos” or any other tool to plot points and found curves.

Submission Date: November 9, 2024 (Saturday – 11:59PM)

Good Luck